# Todo App QA and Debug Report

Application Tested: TodoMVCReactApplication
Link: https://todomvc.com/examples/react/dist/

## Bug Report Table

| # | Title / Summary | Steps to Reproduce | Expected vs Actual | Severity | Suspected Cause |
|---|---|---|---|---|---|
| 1 | No Backup or Data Recovery Mechanism | Create several tasks, clear browser storage, and reopen the application. | Expected: Tasks should be recoverable from a backup system. Actual: All tasks are permanently lost. | High | Data is stored only in local storage with no backup or recovery system. |
| 2 | No User Authentication and Account Management | Open the application and look for sign-up or login functionality. | Expected: Users should be able to create accounts and securely access tasks. Actual: No authentication features are available. | High | The application lacks backend authentication and user management. |
| 3 | No Cloud Storage or Cross-Device Synchronization | Create tasks and open the app on another device or browser. | Expected: Tasks should sync after login. Actual: Tasks are available only in the current browser. | High | Data is stored locally rather than in a centralized database. |
| 4 | No Confirmation Before Deleting Tasks | Create a task and click the delete icon. | Expected: A confirmation dialog should appear. Actual: Task is deleted immediately. | Medium | Delete action is executed directly without confirmation. |
| 5 | No Undo Option After Deletion | Delete a task accidentally. | Expected: An Undo option should allow restoration. Actual: Task is permanently removed. | Medium | Deleted items are removed from state immediately without temporary retention. |

## Root Cause Analysis:-

**No Backup or Data Recovery Mechanism:** The application stores all task data in the browser's local storage instead of using a backend database. Local storage is tied to a specific browser and device, which means the data exists only on that machine. If the user clears browser data, uses incognito private browsing mode, switches to another browser, or closes browser, all tasks are permanently lost. This happens because there is no server side API or database to maintain information outside the user's device. There is also no export, backup, or restore functionality available. Hence, the application is not suitable for production use where users expect their data to be secure and recoverable.

**Solution:** To fix this issue, the application should be integrated with a backend service and store tasks in a centralized database such as MongoDB. Users should authenticate through a sign-up and login system so their data can be associated with individual accounts. The system should also implement regular backups and provide a restore mechanism to recover data in case of accidental loss. This change would significantly improve reliability, scalability, and overall production readiness.